LAB EXPERIMENT RECORD

Set 22: Time Series Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
Date	17 October 2026
Day	Saturday

1. Time Series Line Chart

Aim

To create a time series line chart in R to visualize the trend in monthly sales and label the axes and title the chart.

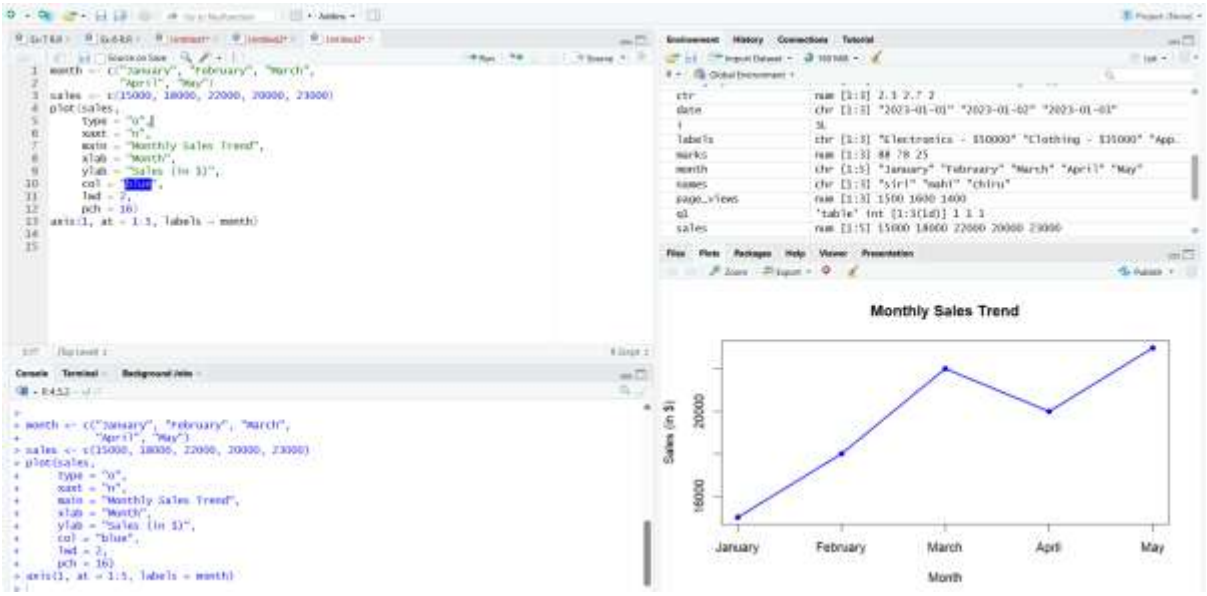
R Code

R

```
month <- c("January", "February", "March",
           "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(sales,
     type = "o",
     xaxt = "n",
     main = "Monthly Sales Trend",
     xlab = "Month",
```

```
    ylab = "Sales (in $)",
    col = "blue",
    lwd = 2,
    pch = 16)
axis(1, at = 1:5, labels = month)
```

Output



A line chart displaying monthly sales from January to May.

Observation

Sales increased from $15,000 in January to $23,000 in May. A slight decrease occurred in April, but sales increased again in May. Overall, the sales trend is upward.

2. Scatter Plot: Advertising Budget and Monthly Sales

Aim

To generate a scatter plot in R to analyze the relationship between advertising budget and **monthly sales and identify whether advertising expenditure is associated with increased sales.**

R Code

R

```
advertising <- c(2000, 3000, 4000, 3500, 5000)

sales <- c(15000, 18000, 22000, 20000, 23000)

plot(advertising, sales,

    main = "Advertising Budget vs Monthly Sales",

    xlab = "Advertising Budget (in $)",

    ylab = "Sales (in $)",

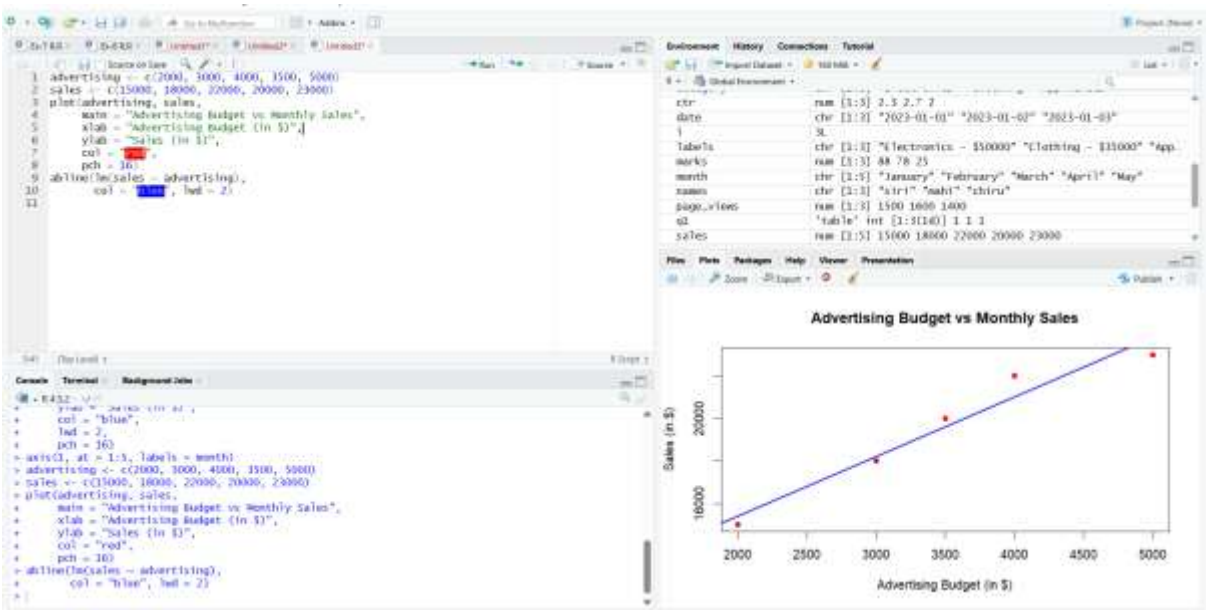
    col = "red",

    pch = 16)

abline(lm(sales ~ advertising),

      col = "blue", lwd = 2)
```

Output



A scatter plot showing the relationship between advertising budget and monthly sales, along with a fitted trend line.

Observation

The scatter plot shows a positive relationship between advertising budget and sales. As the advertising budget increases, sales generally increase. However, the data is illustrative, so this does not establish that advertising caused the increase in sales.

3. Tableau Dashboard

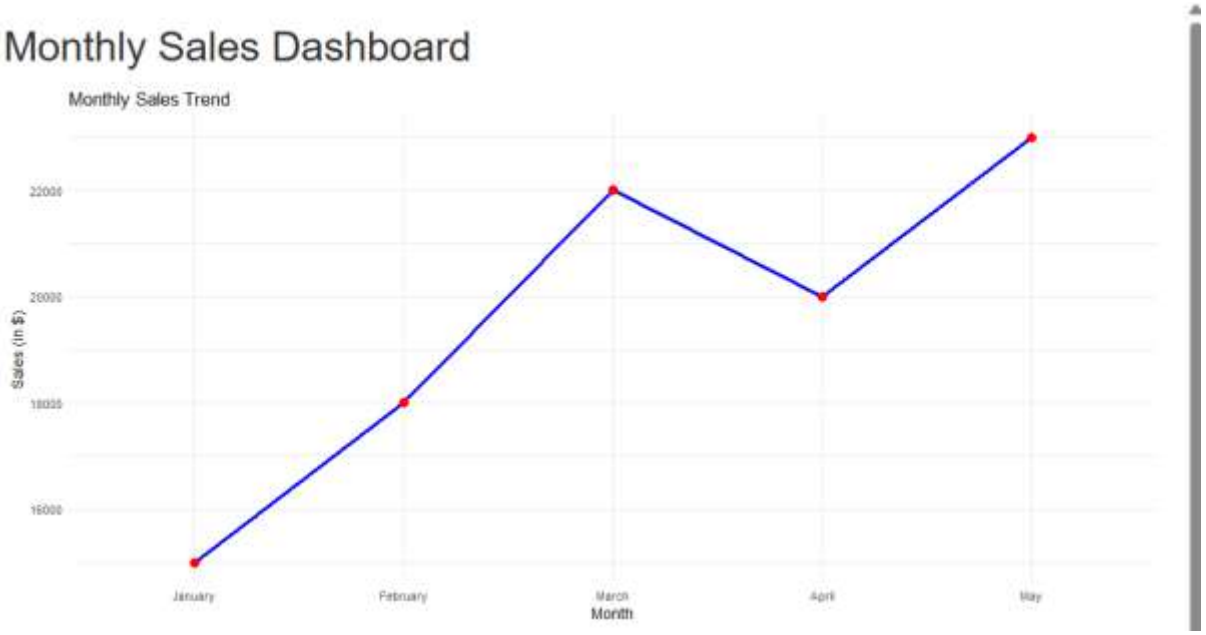
Aim

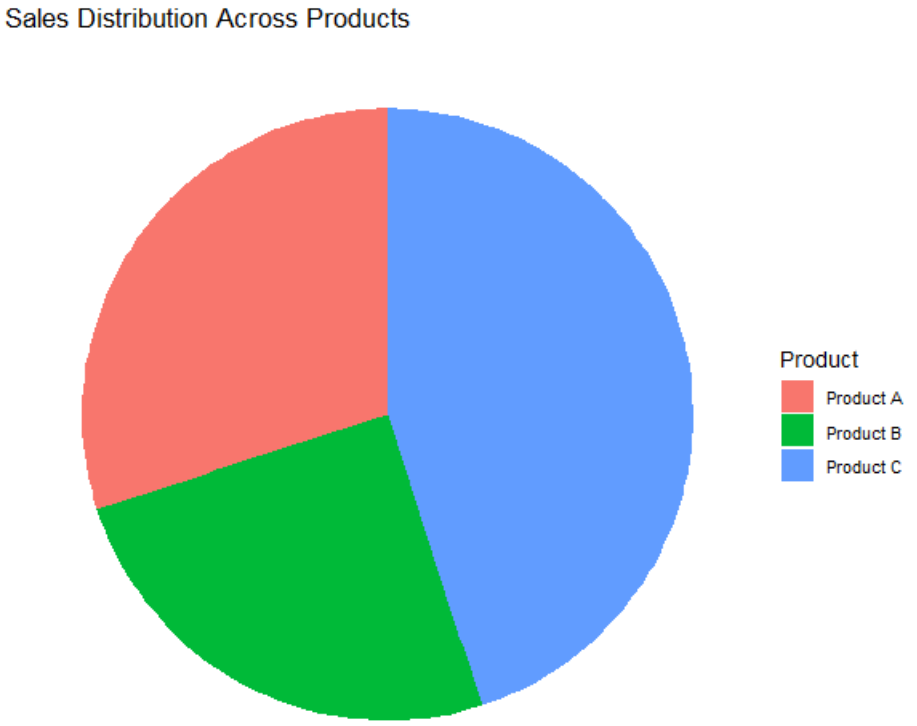
To develop an interactive Tableau dashboard combining a line chart showing the monthly sales trend and a pie chart displaying the distribution of sales across products.

Tableau Procedure

1. Open Tableau and import the monthly sales dataset.
2. Create a line chart using Month and Sales.
3. Arrange the months in chronological order from January to May.
4. Add a suitable chart title and axis labels.
5. Create a separate product dataset containing Product Name and Sales.
6. Select the pie chart option to display sales distribution across products.
7. Assign different colors to the products for easy identification.
8. Select Dashboard and create a new dashboard.
9. Drag the monthly sales line chart into the dashboard.
10. Drag the product sales pie chart into the dashboard.
11. Arrange both charts neatly and add suitable titles.
12. Save the dashboard.

Output





An interactive dashboard displaying the monthly sales trend and the distribution of sales across different products.

Observation

The dashboard helps users understand changes in monthly sales and compare the contribution of different products. Product-wise sales data must be provided separately to create an accurate pie chart.

4. Autocorrelation Plot

Aim

To create an autocorrelation plot in R to examine the relationship between monthly sales values at different time lags and identify possible seasonal patterns.

R Code

R

```
sales <- c(15000, 18000, 22000, 20000, 23000)

sales_ts <- ts(sales,

        start = c(2026, 1),

        frequency = 12)

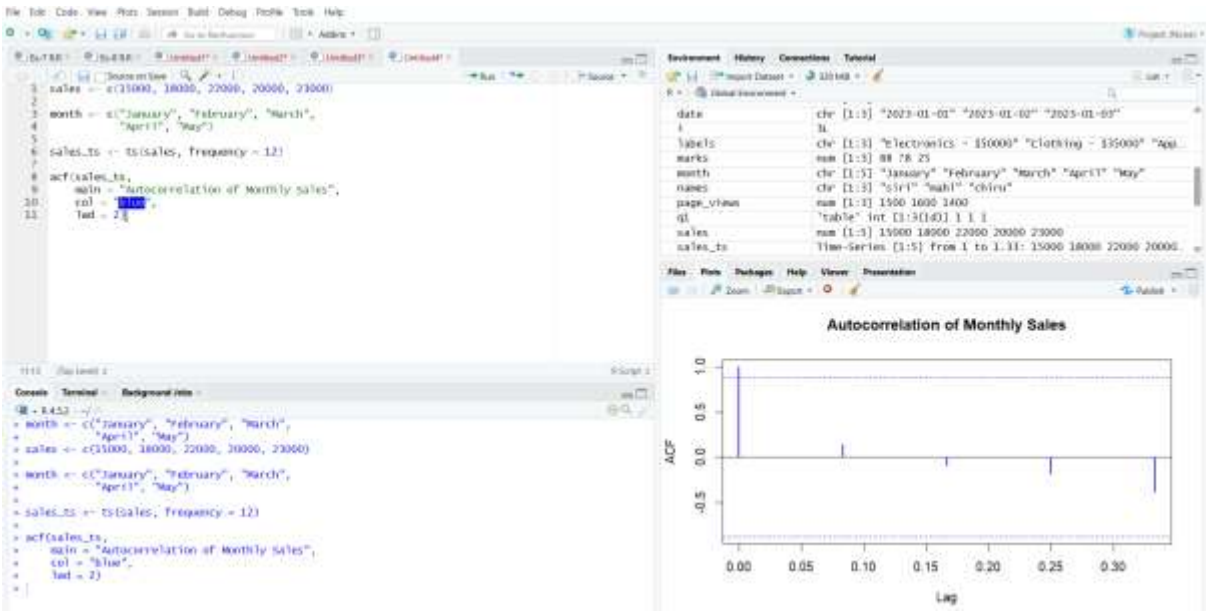
acf(sales_ts,

   main = "Autocorrelation of Monthly Sales",

   col = "blue",

   lwd = 2)
```

Output



An autocorrelation plot displaying the correlation of the sales time series at different lags.

Observation

The autocorrelation plot helps examine how current sales values relate to previous months. However, only five monthly observations are available, which is insufficient to reliably identify seasonality. A longer time series covering multiple years is required for better seasonal analysis.

Result

The monthly sales dataset was analyzed using R through a time series line chart, scatter plot, and autocorrelation plot. A Tableau dashboard was also outlined to combine the sales trend and product-wise sales distribution. The analysis shows an overall increase in monthly sales from January to May.